

ORIGINAL ARTICLE

Characterizing shoulder disorders in Saudi Arabia: A retrospective, single-center study of prevalence, phenotypes, and risk factors

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ABSTRACT

Background: Shoulder disorders are becoming increasingly prevalent worldwide. In this study, the prevalence and predictors of shoulder disorders in Saudi Arabia were discussed, and groups at high risk for such disorders were identified.

Methods: For this single-center retrospective study, the medical records of patients who were diagnosed with shoulder disorders between 2023 and 2025 were retrieved from the physical therapy department in the eastern region of Saudi Arabia. The data analysis included descriptive statistics, classification tree, and binary logistic regression with the aim of assessing associations between shoulder disorder types and risk factors.

Results: The predominate conditions were soft-tissue and degenerative shoulder conditions, with rotator cuff syndrome being the most common (84.5%), followed by capsular-degenerative conditions (12.1%) and then traumatic dislocations (3.4%). Younger individuals were more likely to have traumatic dislocations, whereas middle-aged and older adults were prone to capsular-degenerative disorders, rotator cuff syndrome, and periarthritic disorders. Females had higher percentages of rotator cuff and soft-tissue disorders, whereas males had more traumatic dislocations. Significant independent predictors included sex, nationality, age, body mass index, and residence.

Conclusions: Rotator cuff disorders are the most prevalent shoulder condition. Shoulder disorders are influenced by age, gender, and body mass index. Degenerative shoulder conditions progressively increase with age, and traumatic disorders primarily affect younger patients. Gender-specific patterns are evident, with females being more likely to have rotator cuff and other soft-tissue disorders, whereas males are more prone



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to traumatic dislocations and non-rotator cuff conditions. A higher body mass index increases the risk of non-rotator cuff disorders; compared with obese males, obese females are particularly vulnerable to non-rotator cuff shoulder disorders. (www.actabiomedica.it)

Key words: arthritis, joint dislocations, musculoskeletal diseases, prevalence, risk factors, shoulder

Introduction

Musculoskeletal disorders are not only major health concerns but also significant contributors to disability worldwide (1). Shoulder joint disorders are frequently considered common and debilitating conditions (2). In several countries, the self-reported prevalence of shoulder pain is estimated to be between 20% and 50% (3). In specific populations, such as adolescent athletes, the prevalence can reach as high as 43.5% (4), solidifying its position as the third most common cause of musculoskeletal consultation in primary care (2). In Saudi Arabia, the burden of musculoskeletal disorders is significant, with studies highlighting their high prevalence and disabling nature (5). Musculoskeletal disorders, particularly shoulder joint disorders, are prevalent in Saudi Arabia. A recent study in Taif, a city in Saudi Arabia, revealed an estimated prevalence of shoulder pain among adults of 32.8% (6).

The risk factors for shoulder disorders vary and include work-related, age, and sex determinants. Occupations requiring repetitive arm motions, such as those of health care workers (7), and schoolteachers (8), necessitating substantial lifting, or causing significant biomechanical stress, have been consistently associated with a greater risk of shoulder disorders (9). Other factors, such as working in uncomfortable positions, exposure to vibration, or high force, and handling heavy materials manually, have not only been noted to cause physical stress during work but also increase the risk of shoulder injury (2,9). Personal factors such as poor sleep quality, high physical activity levels, and mental stress have been recognized as predictors of upper quadrant musculoskeletal disorders (10). Furthermore, shoulder injuries are often more common in women than in men (11).

Shoulder issues can impact quality of life and contribute to disability. In Saudi Arabia, studies exploring the rate of shoulder disorders and injury have primarily focused on specific occupations, a specific cohort, or specific geographical regions, which has left a gap in research involving the general presentation of patients seeking rehabilitation. Shoulder pain was reported by 50% of dental students in Jeddah (12) and 31.8% of health care workers in the central region. (13) In northern Saudi Arabia, shoulder disorders were among the most frequent musculoskeletal issues for health care employees, affecting 72.4% of individuals (14). Moreover, regional data have shown similar trends, where 17% of physiotherapists in Taif experienced shoulder issues, (15) 49% of otorhinolaryngology practitioners in the western area experienced shoulder issues (16).

Notably, in none of the studies did the researchers report the affected structure, diagnosis, severity, or risk factors of the shoulder disorder for the patients who were referred for rehabilitation. Therefore, this study aimed to address this gap in research by evaluating the prevalence and categories of shoulder disorders among patients referred to a single physical therapy department in the Eastern Province of Saudi Arabia. The study provides essential epidemiological information and demographic characteristics (e.g., age, sex, and BMI) of groups at high-risk of shoulder disorders. Categorizing the types of shoulder disorders and the affected structures will provide guidance for preventive measures, patient educational programs, and lifestyle modifications. Estimating the prevalence and risk factors of shoulder disorders in Saudi Arabia can provide insight for injury prevention and a better quality of life.

Methods

For this single-center, retrospective observational chart review, secondary data were collected from the medical records of the physical therapy department at King Fahd University Hospital in the Eastern Province of Saudi Arabia. The study design adheres to the STrengthening the Reporting of OBservational studies in Epidemiology Statement, (STROBE) guidelines (17). The data for this study were extracted retrospectively from the hospital's electronic health records system, InterSystems TrakCare® 2024.1 (InterSystems Corporation, Cambridge, MA, USA) on July 15th, 2025. The inclusion criteria for patient records were referral to the physical therapy department of the specified hospital, a diagnosis of a shoulder disorder, and medical records with data from May 22, 2023 (when the InterSystems TrakCare® 2024.1 system was first used), to July 15, 2025. Essential data were collected from the eligible patient records. The data that were extracted included patient demographic information (e.g., age, sex, nationality, weight, height, and residence) and detailed medical history (e.g., present diagnosis, ward, and referring department). Compared with a multicenter study, a single-center study allows for better control over data collection and implementation but is limited in terms of generalizability of the findings (18).

Procedures

Prior to data collection, ethical approval was obtained from the Institutional Review Board of the Imam Abdulrahman Bin Faisal University (IRB-2025-03-0376). In addition, to protect the privacy of the patient records, and the hospital employees, the author signed a confidentiality agreement. Following approval, a meeting was held with the designated data extractor to introduce the study aims, familiarize him with the data collection process, and ensure a standardized approach to data abstraction, which was guided by a specific data sheet tool. Data were extracted from the records of all the patients who were diagnosed with shoulder conditions and consequently referred to the physical therapy departments at the hospital.

Records logged from introduction of the InterSystems TrakCare® medical records system on May 22, 2023, to July 15, 2025, were included in the study. The following information were extracted from patient medical records: demographics, including age, sex, weight, height, nationality, and residential area; medical history, including present diagnosis (e.g., impingement, rotator cuff tear, instability); and referral information: referring department. A standardized case data form was the main data abstraction tool. This form identified all predefined variables in a structured format, which ensured data consistency and minimized variability across the collected records (19). The form was specifically designed to collect the specific information required by the study protocol and was intended for statistical analysis.

Data extractor training and interrater reliability

The data extractor received training to ensure systematic patient identification and precise data abstraction, which was crucial for enhancing the reliability and validity of the data derived from medical records (18). The training included systematic patient identification using clear eligibility criteria. This involved searching the International Classification of Diseases, 10th Revision (ICD-10) codes for the diagnosis of shoulder conditions, reason for admission, and targeted free-text searches within the medical records system to maximize case identification and prevent selection bias (19).

Quality assurance of the data

During the data collection process, quality control measures were applied to maintain a high standard of quality, including double-checking and interrater reliability checks. Entries were double-checked by the primary investigator, where 10% of the abstracted records were randomly selected and double-checked to verify accuracy (20). Furthermore, interrater reliability checks were performed between the extractor and the primary investigator, who was the second rater. Incomplete or missing

data fields were checked. Missing information, such as demographic data was retrieved by reviewing alternative sections of the medical records (21). Data were systematically compiled into a Microsoft Excel spreadsheet. For enhanced data quality control and the prompt identification of problematic variables, Excel's conditional formatting feature was employed to visually highlight cells that were absent or empty (22). Summary tables and bar charts were generated regularly. These visualizations show the proportion of missing values per column or row, enabling rapid identification of variables with data gaps (23). Throughout the process of collecting data retrospectively, patient confidentiality was ensured, and the Excel file provided by the data extractor was fully deidentified and contained no names, contact numbers, addresses, or IDs.

Subgrouping of shoulder disorders:

The shoulder disorders identified by the ICD-10 codes were further regrouped according to the affected structures into clinically meaningful subgroups to facilitate analysis. Following the extraction and validation of the ICD-10 diagnosis codes from the hospital system, patients were classified a priori on the basis of shared pathophysiology, related clinical features, injury profiles, and prior studies (24,25). The Rotator Cuff and Peritendinous group included conditions such as rotator cuff syndrome, bicipital tendinitis, and calcific tendinitis. These conditions were analyzed as a single category because of their close anatomical and functional relationships and overlapping clinical presentation. The rotator cuff tendons are structurally integrated with surrounding peritendinous tissues, particularly the subacromial bursa, such that pathology in one often affects the other (9,26). The second group, the Capsular-Degenerative group, included conditions such as adhesive capsulitis and shoulder arthrosis, which are conditions marked by advancing joint rigidity and degenerative alterations affecting the glenohumeral joint capsule and articular surfaces (27). The third group, the Traumatic-Dislocation group, included traumatic conditions. Prevalence was analyzed by group because of shared symptom profiles.

Data analysis

Data were analyzed using the statistical package for social sciences (IBM SPSS), Ver29. Descriptive statistics were applied, such as frequencies and percentages for categorical variables. Crosstabs and chi-square tests were used to assess the prevalence of shoulder disorders among those referred to physical therapy departments in the selected hospital. A classification tree analysis was conducted to explore the hierarchical structure of predictors associated with the type of shoulder disorder. Binary logistic regression was employed to investigate the association between injury prevalence and the potential predictors, comparing non-rotator cuff disorders with rotator cuff/peritendinous pathologies and estimating odds ratios to describe the direction and magnitude of these relationships. Odds ratios and their 95% confidence intervals were computed, and the threshold for statistical significance was set at $p \leq 0.05$.

Results

Demographic characteristics

Among the 6,938 patients who were referred to the physical therapy department during the study period, 2,523 (36.36%) patients were diagnosed with shoulder disorders. The demographic profile revealed significant associations across nationality, age, and geographical area, whereas sex differences were not statistically significant. The distribution of males and females was comparable, with no significant difference in sex across groups ($p=0.537$). Most participants were Saudi nationals (81.5%), and non-Saudi participants represented 18.5%; such a difference in distribution was statistically significant ($p=0.00$). Saudi Arabian nationals constituted the largest proportion ($n = 2,101$; 81.8%), followed by Yemeni participants ($n = 332$; 13.2%). The remaining nationalities collectively accounting for a smaller proportion of the sample, including Jordanian ($n = 47$; 1.9%), Egyptian ($n = 24$; 1.0%), Palestinian ($n = 22$; 0.9%), German ($n = 12$; 0.5%), Canadian ($n = 9$; 0.4%), Indian ($n = 10$; 0.4%), Syrian ($n = 6$; 0.2%), Sri Lankan ($n = 2$; 0.1%), and Filipino ($n = 2$; 0.1%) participants.

Age was a significant determinant of shoulder disorder distribution ($p=0.00$). The number of patients diagnosed with shoulder disorders ranged from 15 to 81 years, with a mean age of 54.5 years ($SD = \pm 13.20$). Most participants were older age adults, followed by middle-aged adults, elderly adults, and young adults. The distribution of patients diagnosed with shoulder disorders differed significantly among residential areas ($p=0.00$); with most patients residing in Al Khobar (49.1%), followed by Dammam (31.0%), Dhahran (7.3%), and Al Ihsa (4.6%), with the remainder residing in other areas ($n = 202$; 8.0%). The detailed demographic characteristics of the participants are presented in Table 1.

Anthropometric characteristics of patients diagnosed with shoulder disorders

An analysis of demographic data revealed that the mean height of the study sample was 164.21 cm ($SD = \pm 9.13$), ranging from 145 cm to 189.84 cm, while the average body weight was 74.84 kg ($SD = \pm 14.88$), with

a minimum of 40.69 kg and a maximum of 129.42 kg. An analysis of BMI across the study sample revealed that most patients, 972 patients, were classified as overweight (38.5%, CI: 36.6% -40.4%), followed closely by the obese ($n = 802$; 31.8%, CI: 30% -33.6%). However, 678 patients were classified as normal weight (26.9%, CI: 25.2% -28.6%), and only 71 patients (2.8%, CI: 2.2% -3.4%) were classified as underweight. The chi-square goodness-of-fit test ($p = 0.000$) confirmed a statistically significant deviation from an equal distribution across BMI categories, indicating that overweight and obesity were disproportionately prevalent among patients with shoulder disorders. A detailed analysis is given in Table 1.

Patterns of episode presentation and referring departments

An analysis revealed that most patients with shoulder disorders were managed in an outpatient setting ($n = 1,933$; 76.6%, CI: 74.9% -78.3%), followed by inpatient admissions ($n = 522$; 20.7%, CI: 19.1%

Table 1. Demographic characteristics of the patients diagnosed with shoulder disorders (N=2523).

Characteristics		n (%)	P-value
Sex	Male	1246 (49.4%)	0.537
	Female	1277 (50.6%)	
Nationality	Saudi	2057 (81.5%)	0.000
	Non-Saudi	466 (18.5%)	
Age	Adult (15-34)	309 (12.2%)	0.000
	Middle Age (35-50)	434 (17.2%)	
	Older (51-65)	1441 (57.1%)	
	65+	339 (13.5%)	
Residential area (Eastern Province-KSA)	Al Khobar	1238 (49.1%)	0.000
	Dammam	781 (31.0%)	
	Dhahran	183 (7.3%)	
	AlIhsa	119 (4.6%)	
	Other	202 (8.0%)	
BMI	Underweight: < 18.5	71 (2.8%)	0.000
	Normal weight: 18.5 – 24.9	678 (26.9%)	
	Overweight (pre-obese): 25 – 29.9	972 (38.5%)	
	Obese: 30+	802 (31.8%)	

P-values calculated on the basis of the chi-square test for goodness of fit. Bold numbers indicate the difference is significant

-22.3%), with a smaller proportion being handled in emergency departments ($n = 68$; 2.7%, CI: 2.1% -3.3%). The association between episode type and shoulder disorder was statistically significant ($p=0.00$). Information on episode characteristics, including counts, is presented in Table 2. An analysis of referral information (e.g., referring department) revealed a statistically significant distribution of referrals across the sample ($p = 0.00$). Most cases were referred from orthopedics and musculoskeletal departments ($n = 1,695$; 67.2%, CI: 65.4% -69.0%), followed by special or critical care units ($n = 253$; 10%, CI: 8.8% -11.2%),

radiology and intervention services ($n = 188$; 7.4%, CI: 6.4% -8.4%), and general or family medicine clinics ($n = 156$; 6.2%, CI: 5.3% -7.1%). The fewest cases were referred by neurology and neurosurgery departments ($n = 91$; 3.6%), emergency departments ($n = 91$; 3.6%), and other medical specialties ($n = 55$; 2.2%). The data presented in Table 2 and depicted in Figure 1.

Prevalence of shoulder disorders

An analysis of the diagnostic distribution revealed that rotator cuff syndrome was the most prevalent

Table 2. Patterns of episode presentation and referring department among the patients.

Episode Characteristics	Category	n (%)	P-value
Episode Type	Outpatient	1933 (76.6%)	0.000
	Inpatient	522 (20.7%)	
	Emergency	68 (2.7%)	
Episode referring department	Emergency (ED)	85 (3.4%)	0.000
	General / Family Medicine	156 (6.2%)	
	Neurology & Neurosurgery	91(3.6%)	
	Orthopedics & Musculoskeletal	1695 (67.2%)	
	Radiology & Interventional	188 (7.4%)	
	Special / Critical Care	253 (10%)	
	Other Medical Specialties	55 (2.2%)	

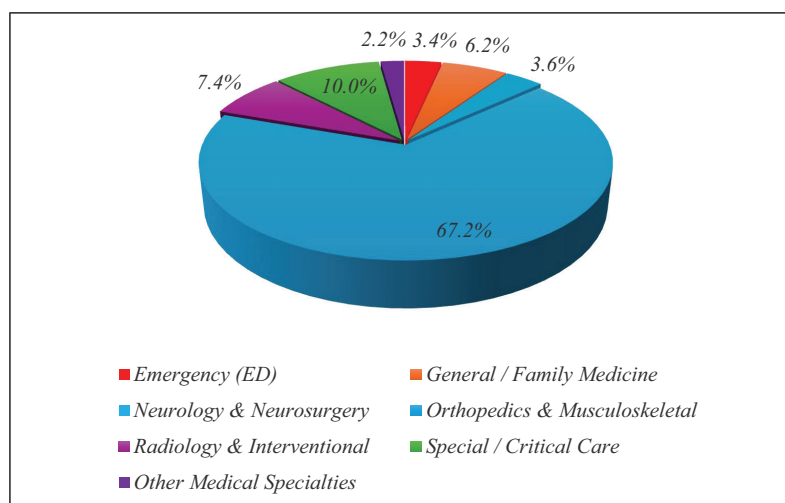


Figure 1. Patterns of care Settings among the patients with shoulder disorders.

shoulder disorder, affecting 1,818 patients (72.1%, CI: 70.3% -73.9%) in the total study population. The second most frequent condition was adhesive capsulitis of the shoulder (n = 261; 10.3%, CI: 9.1%- 11.5%), followed by bicipital tendinitis (n = 236; 9.4%, CI: 8.3%-10.5%). Traumatic shoulder injuries such as dislocation of the shoulder (n = 73; 2.9%) and acromioclavicular dislocation (n = 11; 0.4%) constituted a relatively small proportion (3.2% combined). Similarly, arthrosis of the shoulder region (n = 45; 1.8%) was infrequent. The detailed frequencies are given in Table 3.

The distribution of shoulder disorders across diagnosis categories was highly significant ($p < 0.001$).

Rotator cuff and Peritendinous disorders represented the vast majority of cases, affecting 2,133 patients (84.5%, CI: 83.1% -85.9%). In the Capsular-Degenerative group, which included adhesive capsulitis and shoulder arthrosis, 306 patients were affected (12.1%, CI: 10.8% -13.4%). Finally, in the Traumatic Dislocations group, which included acute shoulder and acromioclavicular dislocations, 84 patients were affected (3.4%, CI: 2.7% -4.1%). Overall, owing to their statistical significance ($p < 0.001$), chronic soft-tissue and degenerative conditions were most common. The subgroups are depicted in Figure 2 and detailed in Table 3.

Table 3. Prevalence of Shoulder Disorders (N=2523).

	Shoulder Disorders	n (%)
Rotator Cuff and Peritendinous	Rotator cuff syndrome	1818 (72.1%)
	Calcific tendinitis of shoulder	79 (3.1%)
	Bicipital tendinitis	236 (9.4%)
Capsular-Degenerative	Adhesive capsulitis of shoulder	261(10.3%)
	Arthrosis, unspecified, shoulder region	45 (1.8%)
Traumatic- Dislocations	Dislocation of shoulder, unspecified	73 (2.9%)
	Dislocation of acromioclavicular joint	11(0.4%)

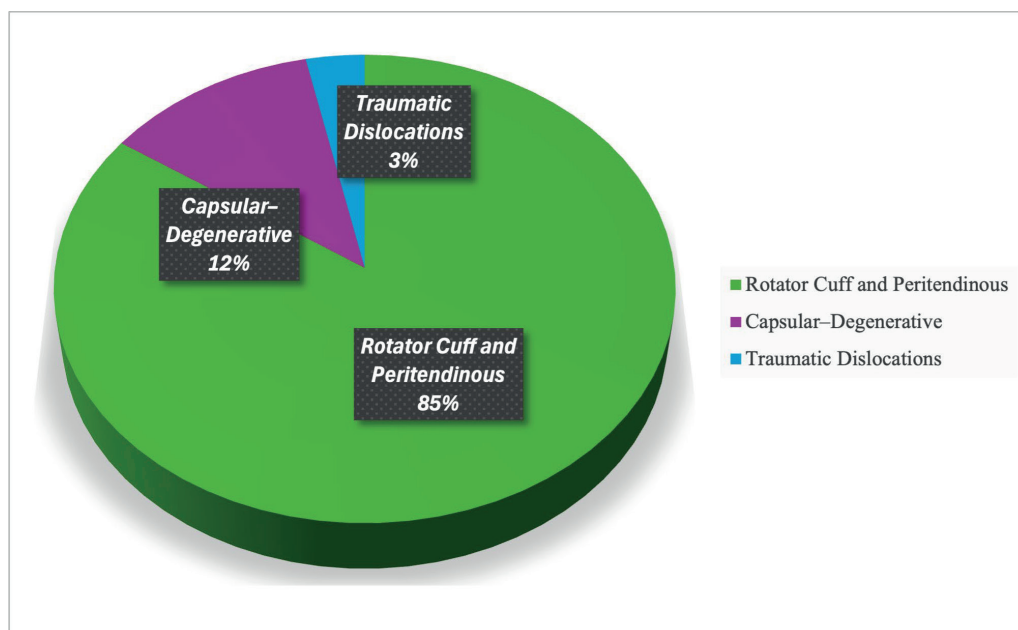


Figure 2. Prevalence of shoulder disorders according to affected structure.

Prevalence of shoulder disorders according to sex

The Pearson chi-square test revealed a statistically significant association ($p = 0.000$) between sex and the type of shoulder disorder. Among male patients ($n = 1,246$), rotator cuff and periarticular soft-tissue disorders were most prevalent, affecting 1,023 individuals ($n = 1,023$; 82.1%, CI: 80.6% -83.6%), followed by capsular-degenerative disorders ($n=157$, 12.3%, CI: 11% -13.6%), while traumatic dislocations were documented in 66 patients ($n=66$, 5.3%, CI: 4.4% -6.2%). Among female patients ($n = 1,277$), rotator cuff and periarticular disorders were even more prevalent, affecting 1,110 individuals (86.9%; $n=1110$; CI: 85.6% -88.2%), followed by capsular-degenerative disorders

(11.7%; $n=149$; CI: 10.4% -13.0%), while traumatic dislocations (1.4%; $n=18$) were notably rare. The types of shoulder disorders prevalent among male and female patients are presented in Table 4.

Prevalence of shoulder disorders according to nationality

The Pearson chi-square test revealed a highly significant association between nationality and the type of shoulder disorder ($p = 0.000$), as presented in Table 4, with Saudi patients experiencing different shoulder disorders than non-Saudi patients do. Among Saudi nationals ($n = 2,057$), the most frequent condition identified was rotator cuff and periarticular soft-tissue disorders, ($n = 1,677$; 81.5%, CI: 80% -83%), followed by

Table 4. Prevalence of shoulder disorders across sex, nationality, age, BMI, and residential area.

	Shoulder Disorders			Total
	Rotator Cuff and Peritendinous n (%)	Capsular-Degenerative n (%)	Traumatic-Dislocations n (%)	
Sex				
Male	1023 (82.1%)	157 (12.6%)	66 (5.3%)	1246 (100%)
Female	1110 (86.9%)	149 (11.7%)	18 (1.4%)	1277 (100%)
Nationality				
Saudi	1677 (81.5%)	296 (14.4%)	84 (4.1%)	2057 (100%)
Non-Saudi	456 (97.9%)	10 (2.1%)	0 (0.0%)	466 (100%)
Age				
Adult	203 (65.7%)	36 (11.7%)	70 (22.7%)	309 (100%)
Middle Age	279 (64.3%)	141 (32.5%)	14 (3.2%)	434 (100%)
Older	1322 (91.7%)	119 (8.3%)	0 (0.0%)	1441(100%)
65+	329 (97.1%)	10 (2.9%)	0 (0.0%)	339 (100%)
BMI				
Underweight	43 (60.6%)	10 (14.1%)	18 (25.4%)	71 (100%)
Normal weight	567 (83.6%)	92 (13.6%)	19 (2.8%)	678 (100%)
Overweight	820 (84.4%)	123 (12.7%)	29 (3.0%)	972 (100%)
Obese	703 (87.7%)	81 (10.1%)	18 (2.2%)	802 (100%)
Residential Area				
AL Khobar	1107 (89.4%)	124 (10.0%)	7 (0.6%)	1238 (100%)
Dammam	593 (75.9%)	143 (18.3%)	45 (5.8%)	781 (100%)
Dhahran	169 (92.3%)	0 (0%)	14 (7.7%)	183 (100%)
AlIhsa	116 (97.5%)	3 (2.5%)	0 (0%)	119 (100%)
Other	148 (73.3%)	36 (17.8%)	18 (8.9%)	202 (100%)

capsular–degenerative disorders in 296 patients (14.4%, 95% CI: 13.0%–15.8%), and traumatic dislocations were reported in 84 patients (4.1%). Conversely, within the non–Saudi patient cohort ($n = 456$), rotator cuff and periarticular disorders constituted the overwhelming majority, accounting for 97.9% (95% CI: 97.3%–98.5%) of the cases. On the other hand, capsular–degenerative disorders were rare and observed in only 10 patients (2.1%), and no traumatic cases were recorded in this group.

Prevalence of shoulder disorders according to age

Age was significantly associated with the type of shoulder disorder ($p < 0.001$), indicating that the type of shoulder disorder is dependent on age. Among younger adults, rotator cuff and periarticular soft-tissue disorders were most prevalent (65.7%, $n = 203$; CI: 63.8%–67.6%), followed by traumatic dislocations (22.7%, $n = 70$; CI: 21.1%–24.3%). However, degenerative and adhesive disorders constituted a smaller proportion in this age group (11.7%, $n = 36$; CI: 10.4%–13.0%). In the middle-aged group, although rotator cuff and periarticular disorders remained the most common conditions (64.3%, 95% CI: 62.4%–66.2%), the prevalence of capsular–degenerative conditions increased substantially ($n = 141$, 32.5%, CI: 30.7%–34.3%), while the prevalence of traumatic cases sharply decreased to 3.2% ($n = 14$). Among older adults, the distribution shifted toward rotator cuff and periarticular disorders ($n = 1322$, 91.7%, CI: 90.6%–92.8%), while the prevalence of capsular–degenerative disorders decreased to 8.3% ($n = 119$), and no cases of traumatic injuries were reported. In the elderly group (65+ years), rotator cuff and periarticular disorders were substantially more common ($n = 329$, 97.1%, CI: 96.4%–97.8%), with minimal representation of other categories. The distribution of shoulder disorder types by age is presented in Table 4.

Prevalence of shoulder disorders according to BMI

The results revealed a significant association between BMI categories and type of shoulder disorder ($p = 0.000$). Among underweight patients ($n = 71$), most

had rotator cuff and periarticular soft-tissue disorders ($n = 43$, 60.6%; CI: 58.7%–62.5%). Traumatic dislocations were observed in 18 patients (25.4%, CI: 23.7%–27.1%), representing the highest percentage of patients with traumatic shoulder disorders across all BMI categories. Both the normal-weight patients ($n = 678$) and the overweight patients ($n = 972$) showed a predominance of rotator cuff and periarticular disorders (83.6%, CI: 82.2%–85% and 84.4%, CI: 83%–85.8% respectively), with relatively low rates of traumatic dislocations (2.8% and 3.0% respectively) and capsular–degenerative conditions (13.6% and 12.7% respectively). The obese group ($n = 802$) had a similar prevalence of rotator cuff disorders (87.7%, $n = 703$; CI: 86.4%–89%). Compared with other BMI categories, obese patients had lower rates of capsular–degenerative disorders (10.1%, $n = 81$; CI: 8.9%–11.3%). The distribution of shoulder disorder types by BMI is presented in Table 4.

Prevalence of shoulder disorders according to residential area

A statistically significant association was observed between geographical area and the type of shoulder disorder, as indicated by the chi-square test ($p = 0.000$) and presented in Table 4. The distribution of shoulder disorders varied across different regions; in Al Khobar ($n = 1,238$), rotator cuff and periarticular soft-tissue disorders were predominant (89.4%, $n = 1107$; CI: 88.2%–90.6%), followed by capsular–degenerative disorders ($n = 124$, 10%; CI: 8.8%–11.2%). In Dammam ($n = 781$), rotator cuff and periarticular disorders were most common (75.9%, $n = 593$; CI: 74.2%–77.6%), and capsular–degenerative conditions accounted for 18.3% of cases, with traumatic dislocations accounting for 5.8%. In Dhahran ($n = 183$), rotator cuff and periarticular disorders were predominant (92.3%, $n = 169$; CI: 91.3%–93.3%), with traumatic dislocations observed in 14 patients (7.7%). Finally, in Al Ihsa ($n = 119$), rotator cuff and periarticular disorders were predominant (97.5%, $n = 116$; CI: 96.9%–98.1%).

Prediction of shoulder disorders

A decision tree analysis was conducted, yielding a statistically significant model (adjusted $p < 0.001$) and

demonstrating a clear partitioning of the sample on the basis of demographic variables. The model revealed age as the strongest predictor (root node) of shoulder disorder classification, followed by nationality, sex, and area of residence. At the first level, age significantly differentiated patients ($\chi^2 = 618.40$, $df = 2$, $p < 0.001$). Among Patients aged 50 years and older (Node 3), 88.1% had rotator cuff and periarticular soft-tissue disorders, whereas 22.3% had traumatic dislocations in the 15–34-year group (Node 1). At the second level, nationality was a significant discriminating factor within the middle-aged group (35–50 years) ($\chi^2 = 42.31$, $df = 1$, $p < 0.001$); within this group, the rate of rotator cuff disorders in Saudi nationals (Node 6) was 76.1% greater than that in non-Saudis (Node 7), with a notable representation of capsular-degenerative disorders (33.3%). Sex emerged as a significant predictor in multiple branches of the tree ($p < 0.001$). Among Saudi adults aged 35–50 years, males (Node 9) had a traumatic dislocation rate of 10.1 %, whereas females (Node 10) had a rotator cuff disorder rate of 84.5%. At lower levels of the model, area of residence was a

significant predictor ($p < 0.001$). Among Saudi males aged 35–50 years, those residing in Dammam (Node 12) had a capsular-degenerative disorder rate of 20.5 %, while those from Al Khobar (Node 11) had a rotator cuff disorder rate of 87.8% (Figure 3).

The Binary logistic regression analysis included sex, nationality, age, BMI, and residential area. To improve model validity, preliminary adjustments were applied to the study variables. The dependent variable was restructured by excluding its smallest diagnostic category, Traumatic–Dislocations, owing to low frequency (<4%), which could compromise model stability and violate the minimum cell size assumptions required for reliable estimation. The independent variables were also modified: age was regrouped into three clinically meaningful categories (15–34, 35–50, and 50+ years) to address sparse data; BMI was collapsed into three categories (underweight, normal weight, and overweight/obese) to minimize multicollinearity and ensure sufficient sample size; and the area of residence was recoded into three categories by merging low-frequency regions. These modifications collectively

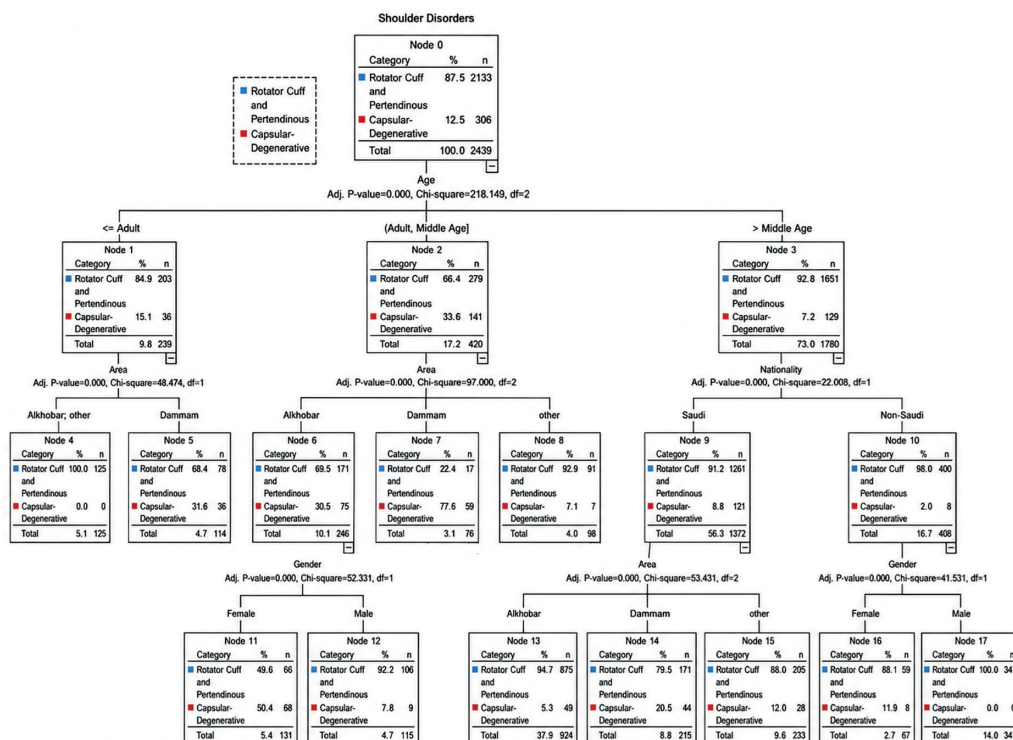


Figure 3. Classification tree of shoulder disorder predictors.

aimed to meet logistic regression prerequisites and enhance model robustness. Binary logistic regression analysis significantly predicted the type of shoulder disorder ($\chi^2(9) = 216.42$; $p < 0.001$), indicating that the predictors reliably distinguished between rotator cuff/peritendinous disorders and other types of shoulder pathology. The model correctly classified 89.2% of cases.

The analysis revealed that sex significantly predicted shoulder disorder type, $B = 0.95$, $SE = 0.38$, $Wald = 6.33$, $p = 0.012$, $OR = 2.59$, 95% CI (1.23, 5.43), indicating that males were 2.6 times more likely than females to present with non-rotator cuff disorders. Nationality was a strong predictor, $B = 3.38$, $SE = 0.36$, $Wald = 87.71$, $p < 0.001$, $OR = 29.29$, 95% CI (14.45, 59.39), demonstrating that compared with non-Saudis, Saudi patients had significantly greater odds of presenting with non-rotator cuff disorders. Age contributed substantially to the model, $Wald = 188.35$, $p < 0.001$. Patients aged 35–50 years were less likely to develop non-rotator cuff disorders than those aged 15–34 years, $B = -0.67$, $p = .010$, $OR = 0.51$, 95% CI (0.31, 0.85), whereas patients aged 50 years and older were substantially more likely to develop non-rotator cuff disorders, $B = 2.13$, $p < 0.001$, $OR = 8.40$, 95% CI (6.08, 11.61).

BMI was also a significant predictor ($Wald = 7.15$, $p = 0.028$). Compared with underweight patients, overweight/obese patients were more likely to present with non-rotator cuff disorders, $B = 0.47$, $p = 0.026$, $OR = 1.59$, 95% CI (1.06, 2.40). Area of residence had a small but significant effect on shoulder disorder type ($Wald = 6.24$, $p = 0.048$). Compared with patients from Al Khobar, those from Dammam had almost twice the odds of developing non-rotator cuff disorders, $B = 0.68$, $p = .041$, $OR = 1.97$, 95% CI (1.03, 3.78). Two interaction effects were statistically significant: residential area $\times$ sex ($Wald = 61.78$, $p < 0.001$). In Al Khobar, males were less likely than females were, to have non-rotator cuff disorders, ($OR = 0.16$, $p < 0.001$), whereas in Dammam, males were more than three times as likely as females were, ($OR = 3.47$, $p = 0.008$). BMI $\times$ sex was also significant ($Wald = 7.90$, $p = 0.019$), with overweight/obese males having lower odds of having non-rotator cuff disorders than obese females did, ($OR = 0.44$, $p = 0.008$). The results of the binary logistic regression are displayed in Table 5.

Discussion

The shoulder joint is a complex structure that allows for a significant degree of mobility in the human body (28). The shoulder joint, while highly mobile, is consequently prone to injury. The current study estimated, categorized, and identified predictors of shoulder disorders in patients referred to a single rehabilitation center in Saudi Arabia. The findings of this study reveal a unique epidemiological profile of shoulder disorders among the examined patient group, highlighting the considerable impact of age, sex, and BMI on the distribution and manifestation of shoulder disorders. The prevalence of chronic soft-tissue and degenerative conditions, especially rotator cuff and peritendinous disorders, confirms that many shoulder disorders are multifactorial and progressive rather than only acute. The age-related distribution of the shoulder disorders revealed a higher prevalence of traumatic injuries in younger individuals and a predominance of degenerative conditions in older individuals. Additionally, there was a clear difference between the sexes in terms of the prevalence of shoulder disorders, with females demonstrating a significantly higher incidence of rotator cuff and generalized soft-tissue pathologies and males exhibiting a higher rate of traumatic dislocations. Strong associations between increased BMI and the incidence and severity of shoulder disorders were observed. Overall, the decision tree model revealed a hierarchical risk pattern, with age being the strongest predictor, followed by nationality, sex, and finally, residential area as a contextual risk factor.

The present study findings indicated that, on average, the distribution of males (49.7%) and females (50.3%) diagnosed with shoulder disorders was similar, with no statistically significant difference ($p=0.79$). However, this finding is notable, as a systematic review indicated that the prevalence of shoulder pain is often higher in women than in men in general populations and high-income countries (3). Interestingly, our study revealed significant sex differences in the prevalence of specific shoulder disorders, which is considered a distinct observation of the current study. The distribution of shoulder disorders by nationality revealed that Saudi nationals represent the primary group affected by shoulder disorders, which is expected, given their

Table 5. Binary logistic regression results for shoulder disorder predictors.

Predictor	B	S.E.	Wald	df	Sig.	Odd Ratio	95% C.I. for Odd Ratio	
							Lower	Upper
Sex- Males	0.95	0.378	6.33	1	0.012	2.60	1.2	5.43
Nationality-Saudi	3.38	0.361	87.71	1	0.000	29.30	14.4	59.40
Area			6.12	2	0.048			
Al Khobar	0.31	0.264	1.33	1	0.249	1.36	0.81	2.28
Dammam	0.68	0.332	4.19	1	0.041	1.97	1.00	3.78
Age			188.35	2	0.000			
Adult	-0.67	0.260	6.71	1	0.010	0.51	0.31	0.85
Middle Age	2.13	0.165	166.54	1	0.000	8.40	6.08	11.61
BMI			7.15	2	0.028			
Underweight	0.85	0.479	3.16	1	0.075	2.34	0.92	5.98
Normal weight	0.47	0.208	5.00	1	0.026	1.60	1.06	2.40
Area * Sex			61.78	2	0.000			
Al Khobar * Males	-1.81	0.450	16.08	1	0.000	0.16	0.07	0.40
Dammam * Males	1.25	0.472	6.95	1	0.008	3.47	1.38	8.76
BMI * Sex			7.9	2	0.019			
Underweight * Males	-1.21	1.003	1.46	1	0.226	0.30	0.042	2.12
Normal weight * Males	-0.83	0.314	7.01	1	0.008	0.44	0.236	0.81
Constant	-6.41	0.436	216.42	1	0.000	0.002		

Abbreviations: B: unstandardized regression coefficient, Wald: Wald chi-square.

majority presence in the local population. The substantial representation of Yemeni patients indicates a notable expatriate subgroup within the sample. Moreover, the smaller proportions reported in other countries suggest that shoulder disorders are not confined to a single nationality but rather occur across diverse demographic groups. The distribution of shoulder disorders by nationality can be attributed to structural and health care system factors, specifically concerning insurance accessibility. In Saudi Arabia, health care access is largely determined by nationality, where citizens receive free public health care as a constitutional right (29). Additionally, public sector employees, including noncitizens, have free access to specified health care facilities (30). In contrast, most private sector expatriates must have health insurance provided by their employers (29). This policy highlights the difference in health care access between Saudi citizens and other

expatriates, which explains the current study's findings, given that the study's data were drawn from a public hospital.

This study revealed that, on average, patients with shoulder disorders were in the middle-to-older adult age group, which reflects the higher prevalence of shoulder disorders among aging patients. The relatively large standard deviation ($SD = \pm 13.20$) indicates substantial variability in age distribution within the sample, which could suggest that shoulder problems affected individuals across a wide age range. The dominance of older adults highlights the age-related degenerative and mechanical nature of shoulder pathologies, which is consistent with the findings of previous studies that link musculoskeletal conditions to advancing age (31,32). This highly significant difference indicates that shoulder disorders are considerably more common among older adults, likely because of

degenerative changes, cumulative occupational strain, and chronic musculoskeletal wear (33). In the present study, the greatest proportion of patients with shoulder disorders who were referred to the rehabilitation center were residents of Khobar, followed by Dammam, and then other areas. The difference in the proportion of referred patients suggests a geographical clustering of patients diagnosed with shoulder disorders, which can be related to the population density or distribution of health care services in the Eastern Province.

The study's sample also revealed a significant prevalence of overweight (38.5%) and obesity (31.8%) among individuals with shoulder disorders. These findings align with recent studies that highlight a significant correlation between increased BMI and many shoulder pathologies (34) and shoulder pain (35). As noted in previous studies, overweight and obese individuals are more likely to develop rotator cuff disorders than other individuals are. Moreover, every 5-unit increase in BMI correlated with a 35% increased likelihood of experiencing a rotator cuff injury (34). These findings clarify that being overweight or obese contributes to the burden and manifestations of shoulder disorders, which is supported by our study findings. In terms of treatment location, the results of the current study revealed that most patients were treated as outpatients (76.6%), which is consistent with research indicating that, because of its typical presentation, chronic shoulder pain is often managed initially in primary care settings before referral to specialists. Furthermore, most shoulder disorder cases are managed in orthopedic and musculoskeletal departments (67.2%), owing to their specialized role. Conversely, acute trauma, severe nerve injuries, and complex cases necessitate immediate or multidisciplinary care and, therefore, are directed to emergency departments (2.7%) or other specialized units, which aligns with the common health care approach in the clinical management of shoulder conditions (36,37).

On the basis of their diagnostic classification grouping in the current study, chronic soft-tissue and degenerative shoulder conditions were predominant. This group of disorders, comprising rotator cuff syndrome, bicipital tendinitis, and calcific tendinitis, accounted for 84.5% of cases. This high prevalence is consistent with the broader epidemiologic trends, and

aligns with the findings of previous studies identifying rotator cuff pathology as a leading cause of shoulder pain and disability, with its prevalence significantly increasing with age (38). The capsular-degenerative category, comprising adhesive capsulitis and shoulder arthrosis, represented 12.1% of the cases. These conditions, characterized by joint stiffness and degenerative alterations, also show an age-related predominance, particularly in middle-aged women (40–60 years), whereas adhesive capsulitis appears to be predominant in older adults (65+ years), which aligns with prior studies where osteoarthritis and previous injuries have been identified as major risk factors (39–41).

Conversely, in the present study, a smaller proportion (3.4%) of shoulder disorders were traumatic dislocations, which is consistent with epidemiological studies revealing that acute traumatic injuries, although common, are less frequent than chronic degenerative conditions (42). Shoulder dislocations are more prevalent among younger, active individuals, with adolescent males being at the highest risk (43). Among older adults and the elderly, the distribution significantly shifted toward rotator cuff and periarthral disorders, accounting for 91.7% and 97.1%, respectively. Moreover, capsular-degenerative disorders declined, and traumatic injuries were absent. The significant prevalence of degenerative soft-tissue pathology in older patients is associated with age-related collagen degeneration, reduced tendon vascularity, and cumulative biomechanical stress (33).

Furthermore, the current study revealed that the rate of traumatic dislocations was higher in males (5.3% vs. 1.4% in females), which is consistent with the findings of previous studies (44,45) that linked them to increased physical activity, occupational load, and sport-related trauma. A greater percentage of females had rotator cuff and periarthral soft-tissue disorders (86.9% vs. 82.1% in males). These results suggest a greater likelihood of degenerative or repetitive strain-related pathologies in females, potentially influenced by hormonal factors affecting connective tissue integrity, such as estrogen and progesterone receptors (46). In addition, biological differences in anatomy, strength, hormones, and musculoskeletal flexibility may contribute to increased shoulder joint laxity and a higher incidence of overuse-related injuries in women

(47,48). The prevalence of capsular-degenerative disorders, such as adhesive capsulitis and shoulder arthrosis, was comparable between sexes, indicating that age-related and metabolic factors may play a universal role in their occurrence (41).

The classification tree analysis presented in this study confirmed that age was the most significant determinant of shoulder disorder type. Younger age (15-34 years) was associated with traumatic dislocations, whereas older age (50 years and above) was associated with rotator cuff and periarticular soft-tissue disorders. Nationality, sex, and area of residence further refined these distinctions. The prevalence of rotator cuff disorders significantly differed within the middle-age group, and sex was associated with traumatic dislocations. The model also highlights the localized influence of the area of residence. Those residing in Dammam had higher rates of capsular-degenerative disorders, while those from Al Khobar had a higher rate of rotator cuff pathology. These findings suggest that occupation and lifestyle have an effect on shoulder morbidity in the Eastern Province (3).

Binary logistic regression was used to evaluate the intensity and direction of these relationships for non-rotator cuff disorders compared to rotator cuff/peritendinous pathologies, complementing the classification tree's understanding. Independent predictors, such as sex, nationality, age, BMI, and area of residence, were statistically significant in the regression analysis. Compared with female patients, male patients had higher odds of non-rotator cuff disorders, and Saudi patients had higher odds than non-Saudis patients did. Middle-aged people were less likely to have non-rotator cuff pathologies than younger people were, whereas older adults were more likely to have such pathologies. An elevated BMI was consistently associated with an increased risk of non-rotator cuff dysfunction, which is consistent with the findings of previous epidemiological reports (3,49). Logistic analysis also revealed important interaction effects. In Dammam, males had a much higher risk of non-rotator cuff disorders than females did, but in Al Khobar, the opposite was true. This might indicate an effect of occupation or lifestyle on shoulder morbidity in the Eastern Province. The study revealed that BMI has a sex-specific effect on shoulder disorders,

with overweight/obese males having a lower likelihood of non-rotator cuff disorders than obese females do. These findings emphasize the importance of analyzing numerous factors together since their combined effect can be more complex than their individual effect, presenting a more complete epidemiological picture. Despite the robustness of the analytical approach, the findings should be interpreted with caution. The observed effect sizes, particularly for nationality and area of residence, may have been influenced by referral bias, whereby certain subgroups (e.g., Saudi nationals) have greater access to specialized orthopedic services. This may have led to an overrepresentation of degenerative shoulder disorders in this group and could partially explain the magnitude of some associations observed in the logistic regression model.

The findings of the current study have substantial clinical implications for personalized diagnostics, treatment, and prevention strategies. Diagnostic evaluations and treatment strategies must be customized according to age- and sex-specific factors. Traumatic injuries and instability should be actively evaluated in younger males. In contrast, attention should be given to chronic degenerative conditions and tendinopathies among middle-aged and older individuals, especially females. The alarming prevalence of overweight and obesity requires the incorporation of BMI status into patient management strategies. Interventions for these groups must account for the increased mechanical loading and its role in chronic degenerative conditions. Addressing obesity is essential because of its association with increased pain and disability and its potential impact on the severity and prognosis of rotator cuff tears. The predominance of peritendinous conditions necessitates a focus on conservative management, such as physiotherapy and therapeutic exercise, for rotator cuff disorders.

The current study has several limitations that necessitate cautious interpretation of the findings. Given its retrospective observational chart review design, the study identified associations, rather than a cause-and-effect relationship, between the factors examined and shoulder disorders. Key unmeasured confounding factors could have affected the observed findings, such as specific occupational factors, detailed physical activity

levels, preexisting comorbidities, dominant limb, or recurrent injuries. Finally, the generalizability of the findings might be limited if the study sample is not fully representative of the broader population with shoulder disorders. Future research should focus on the longitudinal monitoring of individuals to determine the actual causes of shoulder issues, rather than only identifying associated factors. Research should explore a broader spectrum of hidden variables, including comorbidities, occupational demands, daily activities, and stressors, that may influence shoulder health. To ensure the generalizability of the findings, research must encompass a diverse range of individuals and include a greater number of hospitals in different geographic areas.

Conclusions

The current findings indicate a notable prevalence of chronic soft-tissue and degenerative shoulder disorders. As the most prevalent type of shoulder pathology, rotator cuff disorders are predominant among females. Age emerged as the strongest predictor of shoulder pathologies, followed by nationality, sex, and BMI. The multilevel classification model confirmed that degenerative shoulder disorders increase progressively with age, whereas traumatic disorders are predominant in younger patients, and regional factors such as occupational and lifestyle factors influence disease expression. This study conclusively identified sex, age, and BMI as critical demographic and anthropometric factors influencing the prevalence of non-rotator cuff shoulder disorders. Specifically, males, Saudi patients, and overweight/obese individuals exhibit significantly elevated risks for those conditions. These findings highlight the need for tailored screening and intervention strategies that consider distinct individuals' susceptibilities in clinical practice. The model provides clinically meaningful segmentation that can guide personalized prevention strategies and targeted rehabilitation programs for at-risk demographic groups.

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